SEOYOON YUM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) *2022 – Present*
B.S. in Computer Science
Selected coursework: Machine Learning, Deep Learning, Operating Systems, Computer Architecture, Systems Programming, Algorithms.

RESEARCH EXPERIENCE

Undergraduate Researcher, SEAL Lab, KAIST *Sep. 2026 – Present*
Advisor: Prof. Sehoon Kim

- Researching NVFP4 W4A4 quantization and its performance in looped transformer models.
- Optimizing inference kernels to improve inference speedups.

Research Intern, CHEI Lab, UC San Diego *Jun. 2026 – Aug. 2026*
Advisor: Prof. Falko Kuester

- Built a zero-shot anomaly-detection pipeline using the frozen Toto 2.0 time-series foundation model on production Ceph and DigitalPath telemetry.
- Evaluated detection of proxy anomalies and alert-volume tradeoffs; showed that forecast residuals did not reliably predict genuine failures.

PhaseGate – Research Project *Jun. 2026 – Present*
Initiated and led the research; co-authored the paper with Prof. Sehoon Kim (KAIST).

- Built a phase-aware CPU retrieval scheduler for on-device LLMs, achieving 2.0× the throughput of the best feasible fixed policy under the same latency constraint.
- Identified asymmetric CPU–GPU memory contention: four concurrent HNSW searches increased p95 decode latency by 60–61%, versus 5.7–6.9% for prefill, across two Apple M4 systems.

Undergraduate Researcher, SGVR Lab, KAIST *Jan. 2026*
Advisor: Prof. Sung-Eui Yoon

- Evaluated vision-language-action (VLA) policies in simulation under 11 meaning-preserving perturbations; average success fell from 60% to 25%, with structural edits causing the largest drop.

Research Intern, CJ AI Center *August 2023*

- Built a real-time brain–computer interface mapping EEG signals to robotic-arm commands using PyTorch.

PUBLICATIONS

[1] **S. Yum** and S. Kim, "PhaseGate: Phase-Aware CPU Retrieval Scheduling for On-Device LLMs on Unified Memory." NeurIPS 2026 ODI Workshop.

LEADERSHIP

Captain, KAIST American Football Club *Sep. 2025 – Present*

- Lead team practices, game strategy, and intra-university competitions.

Squad Leader and English Interpreter *Jan. 2024 – Jul. 2025*
Republic of Korea Army

- Led eight soldiers and interpreted between Republic of Korea and U.S. military personnel.

TECHNICAL SKILLS

Programming	Python, C
ML systems	PyTorch, LLM inference, low-precision quantization, GPU kernel optimization, performance profiling
Tools	Linux, Git, Bash
Languages	Korean (native), English (professional; TOEFL 110)